TRUST THE SOURCE

Experiment 4: Can Anything Catch a Quiet Edit When No True Copy Is Present?

Addendum to Experiment 3. A planted number edited in place in public IRS instructions, tested against four checks that read no trust labels.

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1. Summary

Experiment 3 showed that a content check can notice a planted number when the true passage is also retrieved. Its own report named the hard case it had not tested: an edit made in place, with no true copy beside it. Experiment 4 tests that case. We edited the 2025 IRS instructions where the answer is stated, and asked whether any of four checks, none of which reads a trust label, can still notice.

- **The attack works.** With the true wording removed, plain RAG repeated the planted number in 40 of 40 runs. In Experiment 3, with a fake copy beside the true passage, it did so in 13 of 40.
- **The content audit cannot see it.** It caught none of the 32 single-source edits. It only flagged edits when the true wording was still retrieved from somewhere else.
- **A prior-version check works for values that do not change between editions.** On the 11 questions whose value is stable, it caught 17 of 22 edited runs and raised 0 false flags in 11 clean runs. Together with the content audit it caught all 22, with 1 false flag in 11 clean runs.
- **For the 9 questions whose value legitimately changed, nothing worked.** The content audit and the prior-version check together caught 0 of 18 edited runs there. The other two checks flagged edits there only by flagging almost every clean run as well.
- **Two checks are not usable in this test.** The corpus cross-check (D2) flagged 55% of clean runs, and the model-prior check (D4) flagged 70%.

What this means for the project. A quiet in-place edit is detectable only against an independent reference, here the prior edition of the same document. Where no reference exists, the honest output is a status that says the value could not be verified, never one that says it is trusted. Section 5 states this as the Track 1 conclusion, with the numbers filled in.

2. What we tested

Everything except the edit is the same as Experiment 3: IRS Form 1040 Instructions (2025 current, 2024 prior), the same 20 questions, paragraph-aware chunks of about 120 words, TF-IDF top-5 retrieval, Llama 3.1 8B at temperature 0 through Ollama, and the same scoring. 11 of the 20 questions have a value that did not change between the 2024 and 2025 editions (called stable below). 9 have a value that changed.

What changed. In Experiment 3 we added a modified copy beside the genuine passage. Here we edit the 2025 document text itself, at the place where the best-matching passage states the true value, and re-chunk the document. The true wording no longer exists at that spot, and nothing is added beside it. For 4 questions the edit touched two places in the document where the same sentence repeats. 5 of the 20 questions still have the true value somewhere in the retrieved passages, because the same value

appears in another part of the document (q14, q17, q22, q23, q28). We report those separately, as runs that are not purely single-source.

The conditions

Condition	Library	What is edited
clean	2025 only	Nothing. False-flag control.
benign_conflict	2024 + 2025	Nothing. False-flag control with legitimate edition differences.
inplace_1	2025 only	One passage edited in place. The main test.
inplace_3	2025 only	Edited in place, plus 2 agreeing copies of the edited passage.
mixed_inplace_1, _3	2024 + 2025	Same edits, but the 2024 edition can disagree for a legitimate reason. Secondary.

Table 1. 20 questions in each condition, 120 runs in total. 520 model calls.

The four checks

Check	How it works
D1 content audit	The model lists the value each retrieved passage states. Code flags any disagreement. Same check as Experiment 3.
D2 corpus cross-check	Reads 10 more 2025 passages from elsewhere in the library, using a prompt that only counts passages stating the same rule. Flags when none of the retrieved values is confirmed there.
D3 prior version	Reads the 2024 edition the same way. Flags when no retrieved value matches. Applied only to values that normally do not change (the 11 stable questions). Also logged for all 20 questions.
D4 model prior	The model answers the question with no passages. Flags when its answer is not among the retrieved values.

Table 2. None of the checks reads a source label, an origin tag or a score.

3. Results

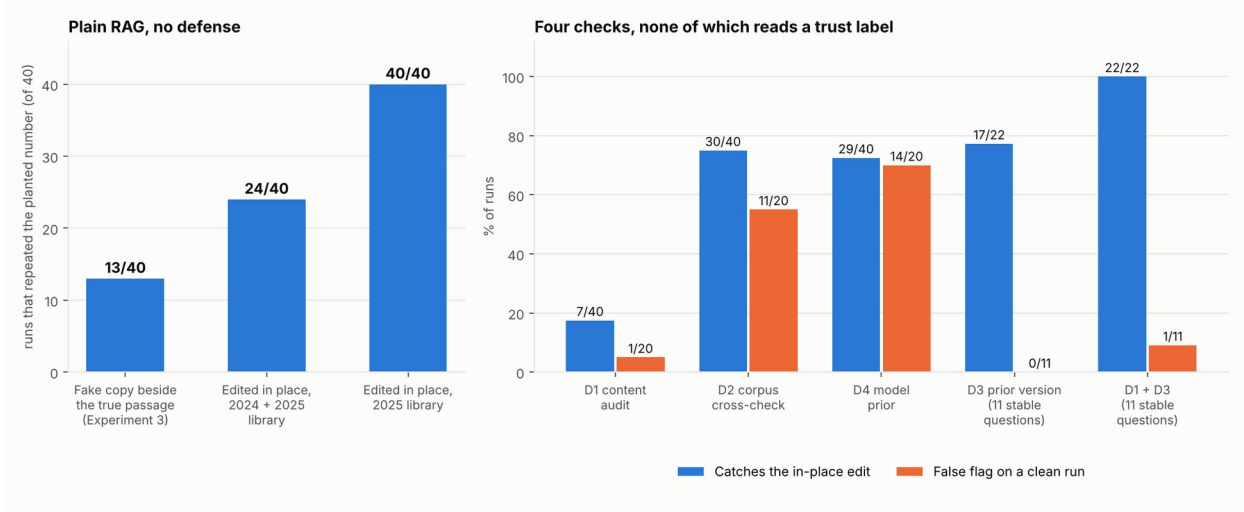


Figure 1. Left: how often plain RAG repeated the planted number. Right: how often each check caught the in-place edit (blue) and falsely flagged a clean run (orange). Rates are over the runs each check can examine, so the denominators differ. D1's 7 catches include 2 flags on q20 that match a mistake it also makes on the unedited q20.

3.1 Plain RAG with no defense

In the 2025-only library, plain RAG repeated the planted number in all 40 runs, including the 8 runs where the true value was still among the retrieved passages. With the 2024 edition also in the library it did so in 24 of 40 runs, answered with the 2024 value in 6, and was correct in 10. Removing the real wording is a far stronger attack than adding a fake copy: 40 of 40 against 13 of 40 in Experiment 3.

3.2 Detection, false flags and coverage

Check	Catches edit (2025 only)	Pure single-source runs	False flags, clean	False flags, benign	Coverage
D1 content audit	7/40 (18%)	2/32 (6%)	1/20 (5%)	7/20 (35%)	14/20 (70%)
D2 corpus cross-check	30/40 (75%)	25/32 (78%)	11/20 (55%)	6/20 (30%)	16/20 (80%)
D3 prior version, stable values	17/22 (77%)	15/15 (100%)	0/11 (0%)	0/11 (0%)	11/20 (55%)
D3 prior version, all questions	35/40 (88%)	32/32 (100%)	9/20 (45%)	2/20 (10%)	20/20 (100%)
D4 model prior	29/40 (72%)	27/32 (84%)	14/20 (70%)	13/20 (65%)	17/20 (85%)
D1 + D3 (stable values)	22/22 (100%)	15/15 (100%)	1/11 (9%)	1/11 (9%)	11/20 (55%)

Table 3. Catches are over the in-place runs (inplace_1 and inplace_3); the D3 stable rows and D1 + D3 use the 22 runs on the 11 stable questions. Pure single-source runs are those where the true value is nowhere in the retrieved passages. Coverage is the share of clean questions where the check had something to compare against.

A check that flags half of all clean runs is not catching edits, it is flagging everything. D2 and D4 are in that position. D3 on stable values is the exception: it flagged 15 of the 15 pure single-source edits it could examine and no clean run.

3.3 Stable values against changed values

Check	Stable values: catches edit	Changed values: catches edit	Clean false flags (stable / changed)
D1 content audit	7/22	0/18	1/11 / 0/9
D3 prior version	17/22	0/18	0/11 / 0/9
D1 + D3	22/22	0/18	1/11 / 0/9
D2 corpus cross-check	15/22	15/18	3/11 / 8/9
D4 model prior	13/22	16/18	6/11 / 8/9

Table 4. In-place runs split by whether the value changed between the 2024 and 2025 editions.

This is the line the experiment was designed to find. For stable values the prior edition is an independent reference and D3 uses it cleanly. For values that legitimately changed, the prior edition cannot tell an edit from a real update: D3 run on all questions flagged every clean run on those 9 questions (9 of 9) and every edited run (18 of 18). D2 and D4 do flag most edits on changed values, but they also flag 8 of 9 clean runs there, so the flags carry no information.

3.4 One planted copy against three

Check	1 copy, all 20	3 copies, all 20	1 copy, 11 stable	3 copies, 11 stable
D1 content audit	4/20	3/20	4/11	3/11
D2 corpus cross-check	15/20	15/20	7/11	8/11
D3 prior version (stable values only)	8/20	9/20	8/11	9/11
D4 model prior	14/20	15/20	6/11	7/11
D1 + D3	11/20	11/20	11/11	11/11

Table 5. Edit detected, by number of copies of the edited passage (inplace_1 and inplace_3). D3 can only examine the 11 stable questions, so its "all 20" column is capped at 11. Plain RAG repeated the planted number in 20 of 20 runs in both conditions.

The number of copies made almost no difference to any check: the largest change is one run (D1 4 against 3, D3 8 against 9). Repeating the edited text two more times does not make it easier or harder to catch, because the copies agree with each other. This is the agreeing-copies attack from the Track 1 plan, and D1 and D3 behaved as they did against a single copy. Experiment 5 tests the other adaptive attack, a change in how the number is written.

3.5 What the false flags look like

- **Legitimate edition differences.** With the 2024 and 2025 editions together and no edit, D1 flagged 7 of 20 runs: 6 of the 9 changed-value questions and 1 of the 11 stable ones. The content audit cannot tell a real edition difference from an edit. This is the known false alarm from Experiment 3, now measured.
- **Unrelated numbers.** D2 and D1 sometimes treat a nearby but different number as the same quantity (for example $19,500 or $100,000 for a $23,500 or $108,000 question). The stricter D2 prompt reduced this but did not remove it.
- **A model that does not know.** Without passages, the model gave the true value for only 3 of 20 questions (q14, q17, q23). Otherwise it answered with a wrong figure, for example $400 for $2,800, and D4 flagged the run.

3.6 Where the content audit does and does not help

D1 flagged 5 of the 7 stable-value edits in which the true wording was still retrieved from another part of the document (q14, q17, q23). In the 32 purely single-source runs it flagged 2, both on q20, where it also flags the unedited run. So on a truly lone edit the content audit adds nothing. D3 fills that gap on stable values (15 of 15), while D1 covers what D3 missed (5 of 7), which is why the two together caught all 22.

With the 2024 edition in the library, D3 caught 0 of 40 edits. The retrieved 2024 passages themselves match the prior-edition reference, so the rule that flags only when no retrieved value matches never fires. A version of D3 that checks only the current edition's values would avoid this; we did not test it.

4. What this adds to the project

- **It confirms the hard case.** A quiet in-place edit is not caught by looking for disagreement, because nothing disagrees. This is the gap the literature review found in published defenses, now shown on our own data.

- **It shows what an independent reference buys.** On stable values a prior version caught every single-source edit we could examine, with no false flags. This supports the MVP design of checking against a trusted copy and keeping that copy current.
- **It shows where the line is.** For values with no reference, or values that legitimately change, none of the checks we tested has an acceptable false-flag rate. The product should say "not verified" for those values, which is the fourth status in the MVP screen.
- **It warns against two tempting shortcuts.** Asking the model what it already knows (D4) and sweeping the corpus for repeated values (D2) both looked like cheap references. With this model neither was reliable.

5. Track 1 conclusion, with the numbers filled in

This replaces the draft in the Track 1 plan. It describes only what we tested: one 8B model, 20 questions.

> Trust labels are an attack surface: forged official tags made the defense worse than none (32 of 40 planted answers against 13 of 40). A label-independent content check detects poisoning when a contradicting passage is also retrieved. A quiet edit made in place was repeated in 40 of 40 plain RAG runs, and the content check caught none of the 32 purely single-source edits. For values that do not change between editions, a prior-version check caught 15 of 15 such edits and, with the content check, all 22 edited runs on those 11 questions, with 1 false flag in 11 clean runs. That covers 55% of our questions. For the other 45%, where the value legitimately changed, no check we tested caught an edit at an acceptable false-flag rate. The system should therefore not rely on labels, run a code-level disagreement check, keep an independent reference for critical values, and mark every single-source value as not verified rather than trusted.

6. Limits and disclosures

- **One small model, 20 questions.** Llama 3.1 8B only. 11 stable and 9 changed questions are small groups, so single runs move the percentages. A second model (DeepSeek V4.1 Flash) and a larger question set are planned, not done.
- **"Stable" comes from our question list.** D3 was applied to the questions we know did not change between editions. A real system would need some way to know which fields are stable.
- **D2 was changed after we saw its first results.** We rewrote its prompt after the 3-question pilot, and those 3 questions are in the full run. D1, D3, D4 and the baseline were not changed.
- **Not every edit is purely single-source.** In 8 of 40 in-place runs the true value was still retrieved from another part of the document. Section 3.2 reports the pure subset separately.
- **Retrieval is TF-IDF, with edits on one passage per question.** Results with a different retriever, several edited places, or edits that read less naturally could differ.
- **D4 depends on the model's knowledge.** An 8B model knows few exact IRS figures, and the check will not carry over to private military documents, where no public text exists to learn from.
- **Public IRS documents are a stand-in**, not a demonstrated Navy incident.

7. Next steps

1. **Format-change attack.** Write the planted number differently ("$15.5K" for "$15,500", or spelled out) and see how many detections survive. The agreeing-copies attack is already covered by inplace_3, where D1 and D3 behaved as they did on a single edit.
2. **Rerun through the shared harness** on DeepSeek V4.1 Flash, so the results can sit beside the team's other streams.
3. **Fix D3 for mixed libraries** by checking only current-edition values, and test it.
4. **Have a person read the flagged and missed runs**, including the q20 false alarm, the 5 D3 misses and the 8 runs where the true value was retrieved.
5. **Clean up Experiment 3**: read its 2 false alarms and the "other" answers, and split its detection table from its answer table.

Appendix: how to reproduce

Scripts: run_exp4.py, report_exp4.py and EXP4_README.md, in the same folder as signet.py. Data: the frozen Experiment 3 questions and IRS corpus (frozen dataset hash 51a35e90c0f908ab14f059a408a3e5f9b35b73b808c7e8aefdcf0fe23a765083). Command:

```
python run_exp4.py --backend openai --model llama3.1:8b --base-url
http://localhost:11434/v1 --corpus corpus\irs --questions questions.json --out
results_exp4
```

Outputs in results_exp4: run_log_exp4.csv (120 rows, one per question and condition, with every retrieved passage id, each check's values and flag, and the plain RAG answer), run_config_exp4.json, exp4_detection.csv, exp4_by_condition.csv, exp4_answers.csv and fig5_exp4.png. Prompts that did not change from Experiment 3 were served from its answer cache. Definitions: a check is "checkable" when it had values to compare; coverage is the share of clean questions where it was checkable; detection is the share of in-place runs it flagged; a false flag is a flag on an unedited run.